Chatfield Quantitative Research: Week 1 Summary

Role: Quant 3, Python Implementation and Forecast Evaluation

I developed a Python prototype to establish the forecasting and evaluation workflow for our project, “Can Online Job Postings Improve Forecasts of UK Labour-Market Vacancies?” The script loads a monthly series, separates training and test observations, estimates ARIMA, produces a naive benchmark and saves forecasts, accuracy scores and a comparison plot.

The example dataset contains 120 synthetic monthly observations from January 2015 to December 2024. It starts at 100, with each subsequent value calculated by adding a simulated monthly change. Each change equals 0.6 times the previous change plus an independent standard normal shock, starting from an initial change of zero. A fixed random seed of 42 makes the data reproducible. This creates persistence in monthly changes without requiring economic data collection. The values have arbitrary units and do not represent UK vacancies.

I implemented ARIMA(1,1,1) using statsmodels, with one autoregressive lag, first differencing and one moving-average lag, without drift. This was a simple initial specification for demonstrating the workflow, not the outcome of model selection. The first 108 observations, January 2015 to December 2023, were used for estimation; the final 12 months were reserved for evaluation.

Both methods forecast the full test period from the same December 2023 origin. ARIMA was fitted once, while the naive benchmark repeated the final training value across all 12 months. Neither method received actual test observations during forecasting. This ensures a consistent comparison, although the resulting scores combine horizons one to twelve rather than representing repeated one-month-ahead forecasts.

Model	MAE	RMSE
Naive	4.8680	5.5446
ARIMA(1,1,1)	3.2728	4.0241

MAE measures the average absolute error, while RMSE places greater weight on larger errors. ARIMA reduced MAE by approximately 32.8% and RMSE by 27.4% relative to the benchmark. However, the plot shows that both forecasts missed much of the subsequent decline. ARIMA's lower errors therefore indicate relative improvement on this holdout, not reliable prediction of every movement. The synthetic process also deliberately contains autoregressive structure. These results cannot establish performance on UK vacancies or the value of Indeed indicators.

One technical issue was inconsistent CSV column naming between data generation and loading, which was corrected. A subsequent check reproduced the saved forecasts and scores. Remaining limitations include a single synthetic series and forecast origin, no parameter optimisation or rolling backtesting, and limited input validation and convergence checking.

The code separates loading, modelling, evaluation and plotting into functions. To adapt it, I will replace the synthetic input with the agreed vacancy series and confirm frequency, date labels, units and transformations with the other analysts. Later work will add Indeed predictors and rolling-origin evaluation. This will require particular care over publication timing and which predictor values are available at each forecast origin, so that future information does not enter the comparison.